

User involvement and participatory design

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Content of the lecture

Session 1 (13.15-14.00)

- The concept of user involvement
- User-centred design vs. participatory design vs. co-design
- Examples, methods and tools of participatory design and co-design

Session 2 (14.15-15.00)

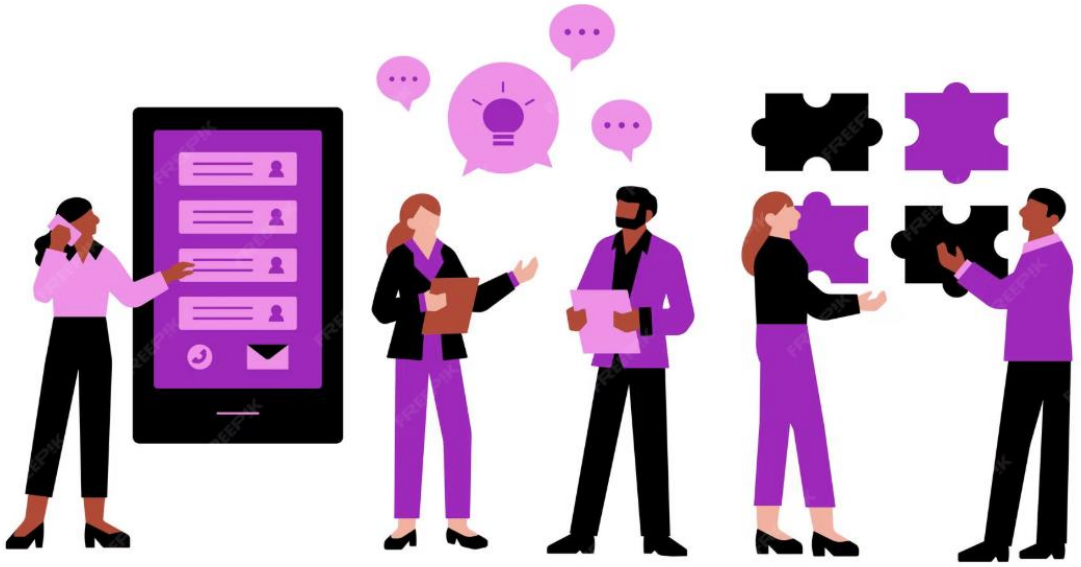
- Critical perspectives on user involvement
- Social Learning Perspective
- Levels of user involvement
- When to participate? Why and whom to involve?



User involvement

- One term, many definitions:

- `focus on users' ,
- contacting with system users',
- `participation of users



"User involvement can be seen to be a general term describing direct contact with users and covering many approaches."
(Kujala 2003, p.1)

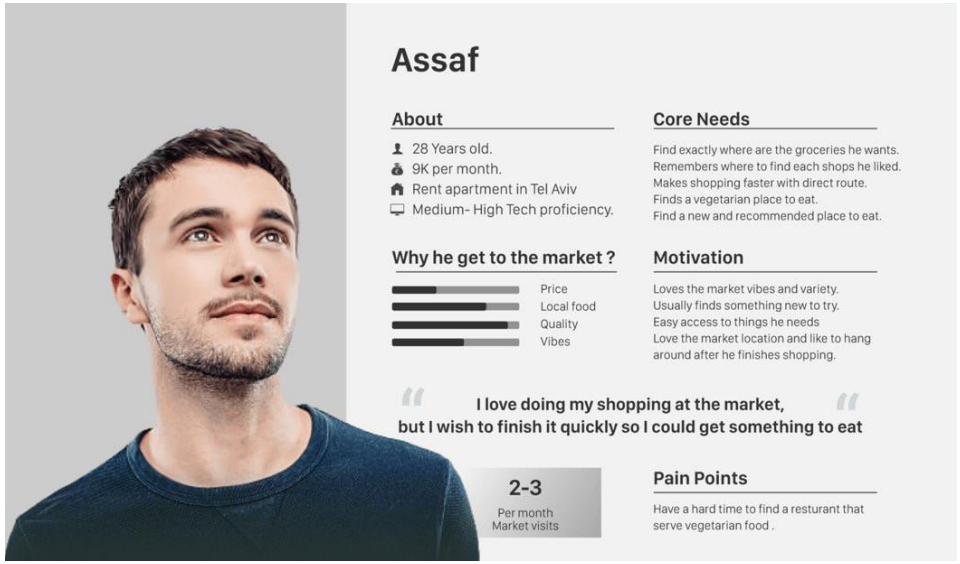
Includes PD, UCD, Co-Design, UX Design, Usability testing, Prototype testing...

User involvement - Definition

- Design approach to understand and integrate user needs in the design process
- Co-producing interfaces, audio, visual or technical elements together **with users** (rather than for users)
- Users can produce ”alternative options” or even reject designers’ intentions
- User involvement is expected to improve the adaptability of technical system to user’s needs, skills and preferences, and different contextual and environmental factors.



For users – or with users?





Co-designing technology– or activity?

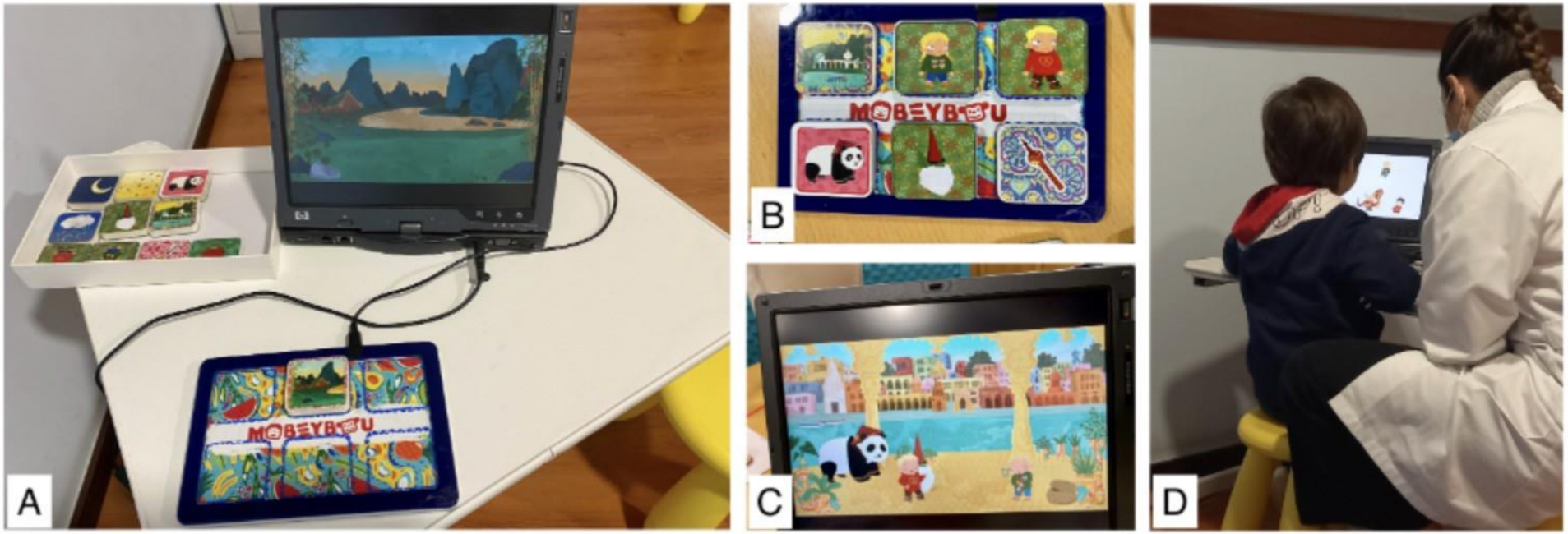


Fig. 1. **A:** The platform with smart tangibles with a screen with the visual representation of the story content connected with a cable to the tangible board where the child puts the blocks on. **B:** An example co-designed activity, Magical Objects as a Strategy Sequence: 1 landscape + 2 children + 1 antagonist + 1 animal and 1 magical object on the tangible platform. **C:** Example output of co-design session with the corresponding digital story projection on the screen. **D:** The therapist is independently using the system within a therapy session of a child with ASD.

Guneysu, A., Reis, H., Kuoppamäki, S., Sylla, C. (2024) Enhancing Autism Therapy through Smart Tangible-Based Digital Storytelling: Co-Design of Activities and Feasibility Study. In *Interaction Design and Children (IDC '24)*. Association for Computing Machinery, New York, NY, USA.

How do users matter?

1. What users *do*

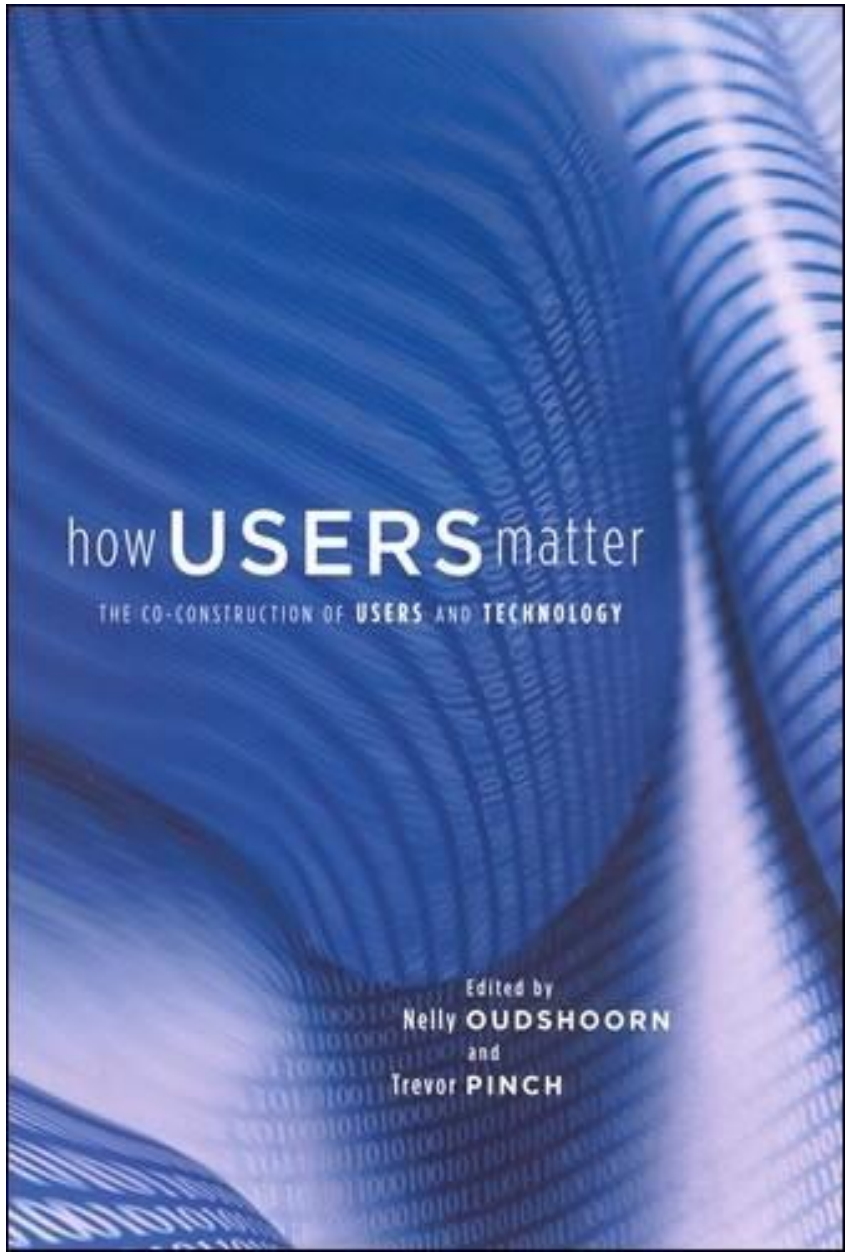
- Users have a capacity to shape technologies
- Users as designers, producers, marketers, distributors
- Users as agents of technological change. Refusing to use a technology or using it in unintended ways is a powerful form of user agency

2. Theories

- SCOT (Social Construction of Technology) approach
- Semiotic approaches: configuring the user
- Cultural and media studies: Consumption and domestication

3. Critical perspectives

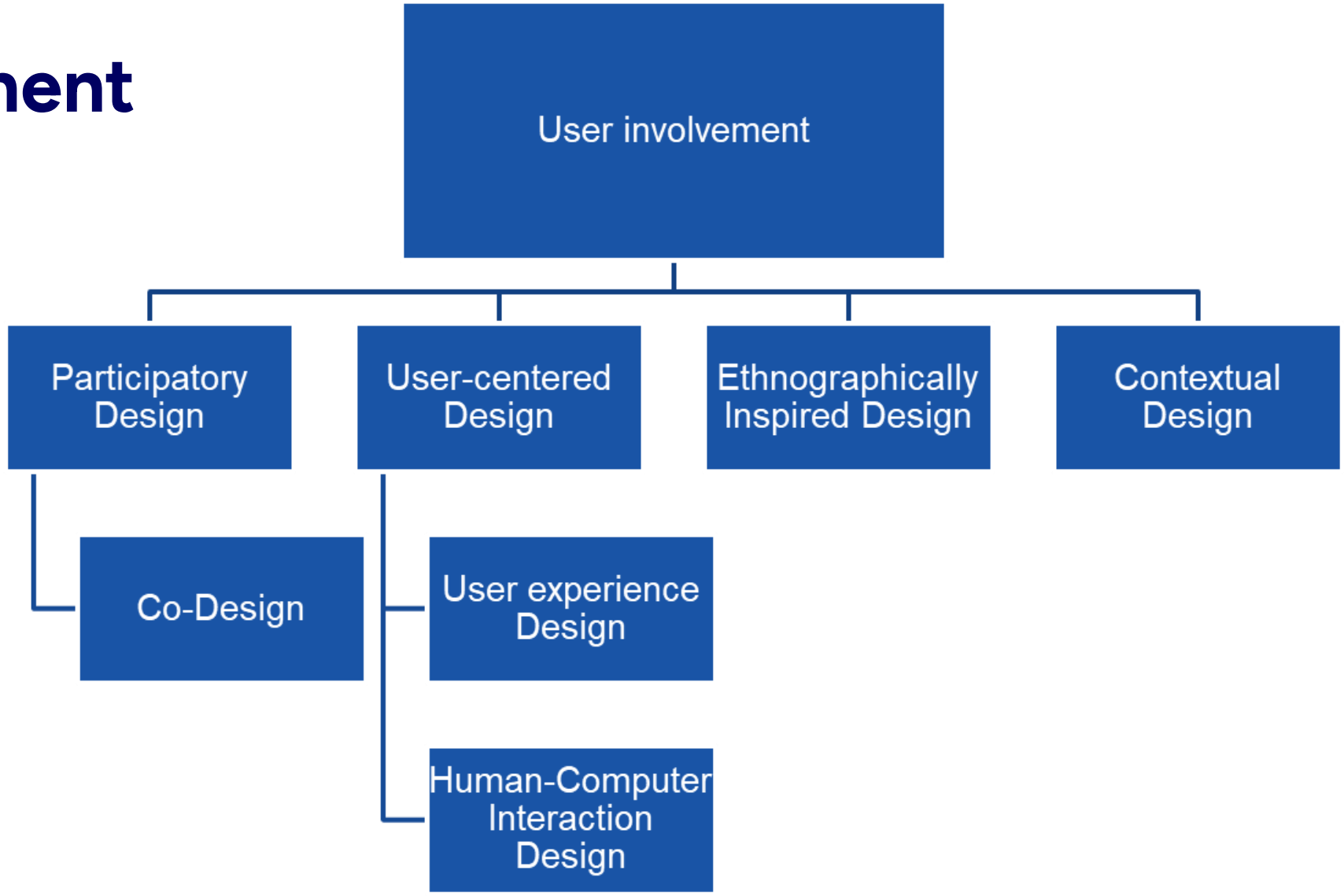
- Feminist approaches: Diversity and power
  - Technologies may have problematic consequences for women
  - Absence of women in historical development of technology
- Issues of exclusion, power, inequality: Focusing on users rather than machines would result in more inclusive technology design



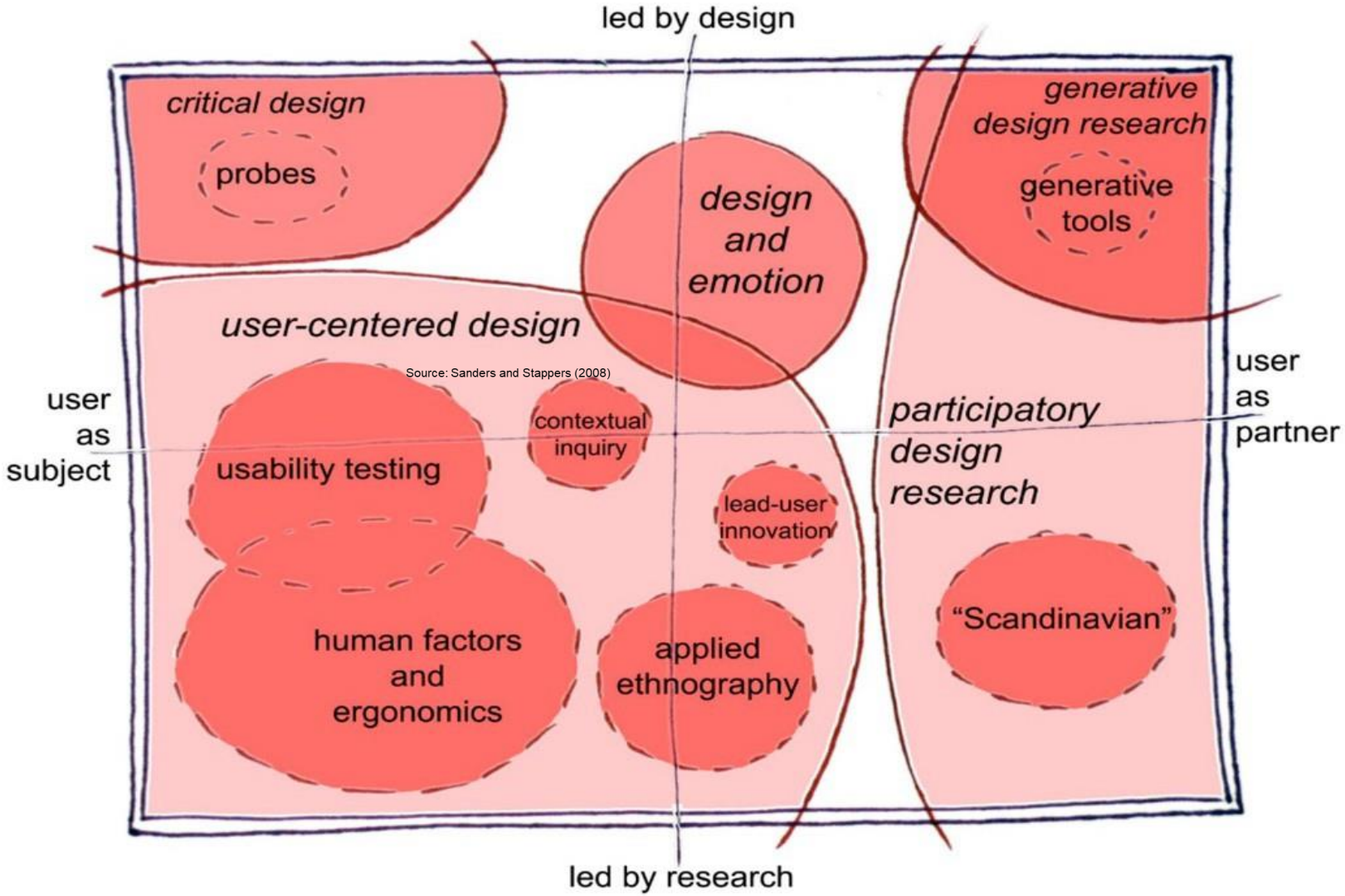
User's Perspective vs. Designer's Intention



User involvement



User involvement: Relationship among approaches



Source: Sanders and Stappers (2008)

User involvement: Differences among approaches

	Participatory design	User-centered design	Ethnographically inspired design	Contextual design
Focus	• Demographic participation • Empowerment	• Usability • 'Product' meets 'needs'	• Social aspects of user	• Context of user
Methods	• Workshops • Rapid prototyping	• Need analysis • Personas • Usability testing	• Observation • Video analysis	• Field interviews

Participatory design vs. user-centered design





- User as **partner** vs. User as **subject**
- Design for people's **purposes** vs. Design of categories of **products**
- **Designing the activity vs. product**

The traditional design disciplines focus on the **designing of** "products".	. . . . . while the emerging design disciplines focus on **designing for** a purpose
visual communication design interior space design product design information design architecture planning	design for experiencing design for emotion design for interacting design for sustainability design for serving design for transforming

Source: Sanders and Stappers (2008)

Participatory design: Fundamental aspects



Politics - People who are affected by a decision should have an opportunity to influence it



People - People play critical roles in design by being experts in their own lives



Context - The use situation is the fundamental starting point for the design process



Methods - Methods are means for users to gain influence in design processes



Product - The goal of participation is to design alternatives, improving quality of life

Findings from Halskov and Hansen (2015), based on a review of core PD literature

Characteristics	User-centered Design	Participatory/Co-Design	Human-Centered Design	Participatory Research
How are the end-users of the technology being conceptualised	User as Informants (providing feedback) User as subject	User as Partner, active or full participant, co-designer	Humans Active	equal, active full participant, co-researcher
Key focus, overarching characteristic	**Usability**	**Collaboration**	**Empathy Meaning-making**	**Empowerment**
Working principles or values	Improving the understanding of user and task requirements	Democracy Interactive two-way relationship between designer and user	Gaining a clear understanding of how users, interact with their environment	Addressing issues which really matter to disabled people and ultimately leads to improved lives for them
All or some stages in the process	Typically, early in the development cycle (but not necessarily in the initial idea stage) but can be throughout	Throughout	Throughout	Throughout (including initial idea stage and final analysis dissemination stage)
Processes	Iterative Design, Empirical Measurement	Iterative	Reflective Evaluative Iterative	Reflective, Co-operative
Methods, Tools and techniques	Task analysis, needs analysis, Usability testing, heuristic evaluation, prototyping (lo-tech, rapid)	Ethnographic methods, Mock-ups, Games, role play, acting, Workshops; Diaries, scenarios	Consulting data-sets; ethnographic interviews and observations; focus groups, role-playing; think-aloud	Life-stories Individual or group interviews
Goals and/or motivations	**A better (more usable) product**	**Better quality of life (through use of end-product) Better end-product**	**A better usable product Improved quality of life for users**	**Improved living conditions**



Figure 2: Participants drew their menstrual cycles and shared menstrual experiences (left), which were documented in a menstrual cycle "artwork" (right).



Figure 4: The participants' five design concepts of technologies for menstrual cycle experiences: Värme Plagget (left), The Calm-inator (middle), U-formen, Korven, and Transformer (right).



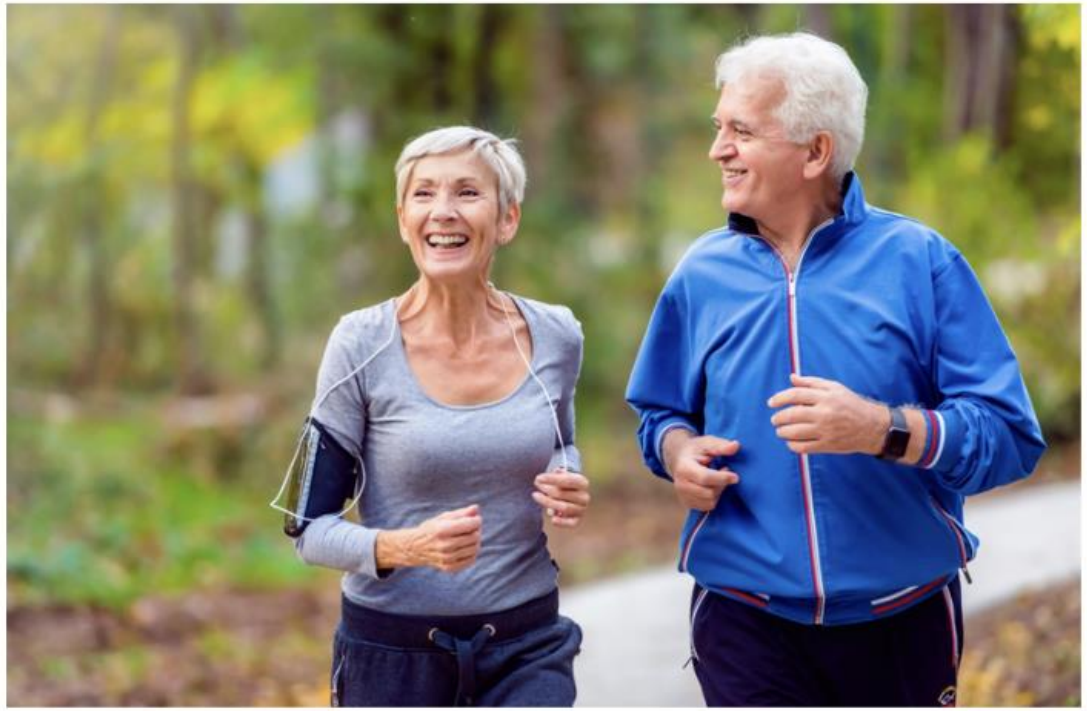
Marie Louise Juul Søndergaard et al. Designing Menstrual Technologies with Adolescents. (2021) https://doi-org.focus.lib.kth.se/10.1145/3411764.3445471

Discussion 1 (15 mins)

- What are the differences between various approaches in user involvement?
- What approaches are you familiar with?
- What is applicable to your field?
- Document your discussions and insights in Google Doc: https://bit.ly/3LtAQYH



Example: Age-Wear Project



- In what physical activities and contexts do older adults use or are interested in using wearable technologies?
- **What wearable design features encourage long-term adherence to exercise routines and device use?**
- How can different modalities (auditory, haptic) and feedback systems enhance the heart-step synchronisation?

Phase 1
User Study

1) Interviews and questionnaires

- Habits and routines
- View on technologies
- Wearable use
- self-perceived health

2) Home study

- 3-week usage of the device in the home environment

3) Post-interviews and questionnaires

- Usability
- User experience

Phase 2
Co-design (workshops)

Topic 1: Materials & Aesthetics (with older adults)

- Placement
- Shape
- Colour

Topic 2: Movement Design (with movement experts)

- Inclusive movements
- Physical activity promotion

Topic 3: Modality Preferences (with older adults)

- Sound
- Music
- Haptic

Phase 3
Impact evaluation

Development of a new device.

1-3 months home study with the developed device.



Understanding user's needs



Participants:

- 18 older adults (aged 73±4, 13 female and 5 male)
- Recruited from community centres and senior activity groups.

Pre-interview & survey on physical activity, health, and use of technology

Three-week home use of WRF (at least 3 times a week, 30 mins for each time)

Post-interview & follow-up survey on user experience and usability

Lu TY, Che J, Rosato A, Dual SA, Kuoppamäki S
Lived Experiences of Older Adults Using Wearables with Real-Time Feedback: A Phenomenological Study https://preprints.jmir.org/preprint/71509

Older Adults & Wearable Devices

- **Social Technology Adoption**
  - Relied on "warm experts" (family/professionals) for setup and data interpretation
  - Technology adoption is socially embedded, not individual choice
- **Context-Dependent Needs for Rhythmic Activity**
  - Effectiveness depends on intersection of bodily capacity, environmental demands, and tech requirements
- **Technology as Relationship Mediator**
  - Devices mediated consciousness-movement connection
  - Could fragment OR enhance embodied experience
  - Temporal Engagement duration is critical for adherence

Steps in the co-design process



Challenge: 'Participants may do not have any experience on robot technologies, or they may do not see themselves as designers'

- **1) Group Discussion:**
- Focus group discussion about expectations (Design scenarios)
- Individual interaction with wearables (open-domain dialogue for 7-minutes)

- **2) Participatory design workshop**
- Introducing the goals
- Design activites
- Individual questionnaire about the workshop – reflective designers

Example: Design Scenario













- What kind of activity would you like to use the wearables?
- What would you like that the wearable to be placed on your body?
- What would you like to the wearable to measure?
- Do you think your health data would useful for you? Why, why not?

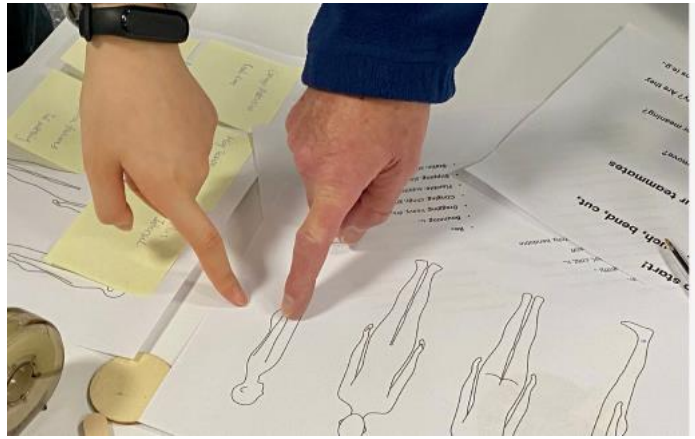
Ask ***How/why questions!***

Guide the participants to work with you








- Adapt the appropriate design methods and design probes

Making design choices with participants



Critical perspective on user involvement



Traditional Product Development, https://buzzclan.com/digital-transformation/product-development/

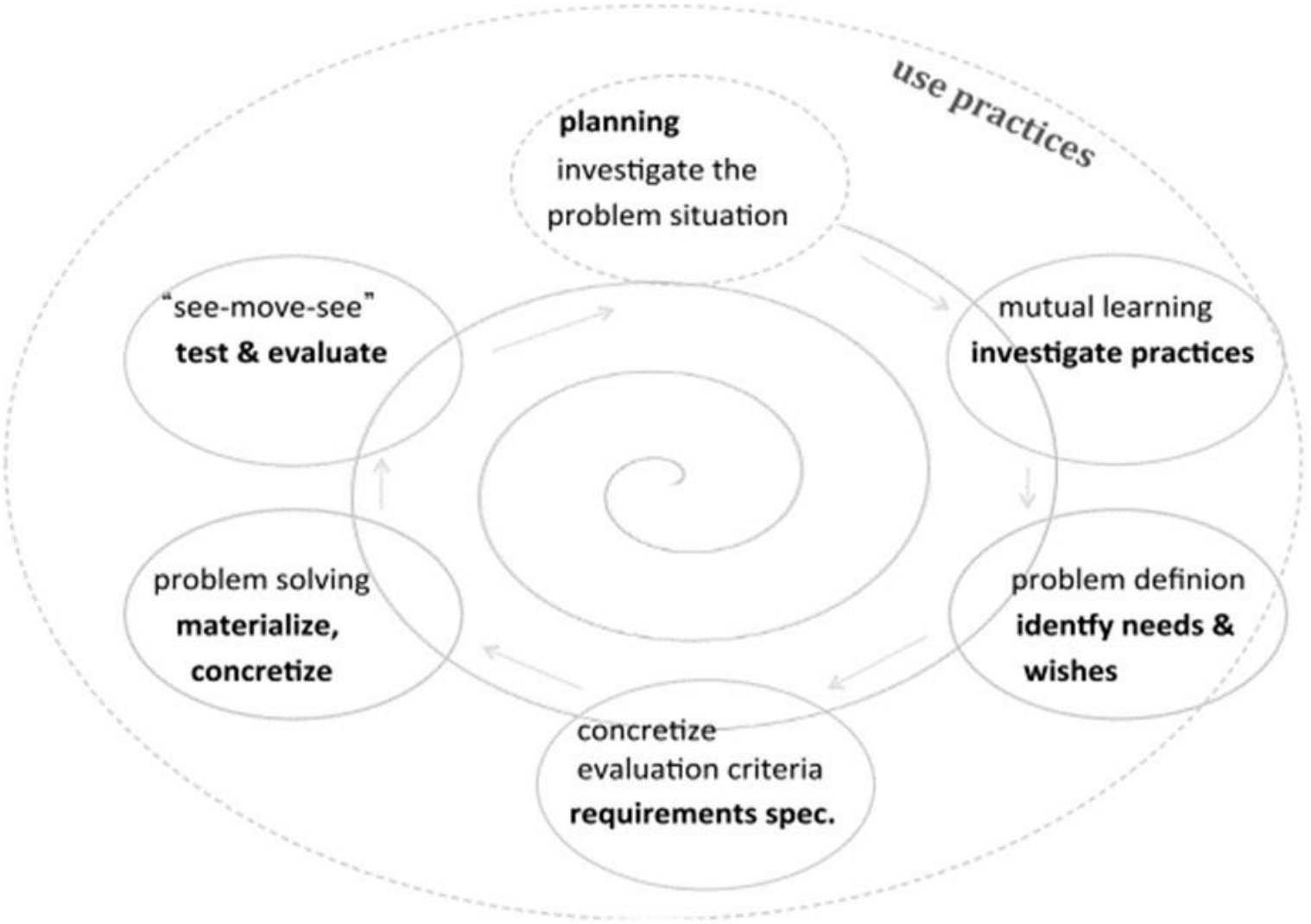
Critical perspective on user involvement

The social learning perspective (Stewart and Williams 2005)

- Against design fallacy of accumulating ever more information about current user requirements.
- Innovation continues from design of artefact to its implementation and use
- User needs and requirements evolve as users incorporate technologies into their everyday environment
- Feedback of user insights

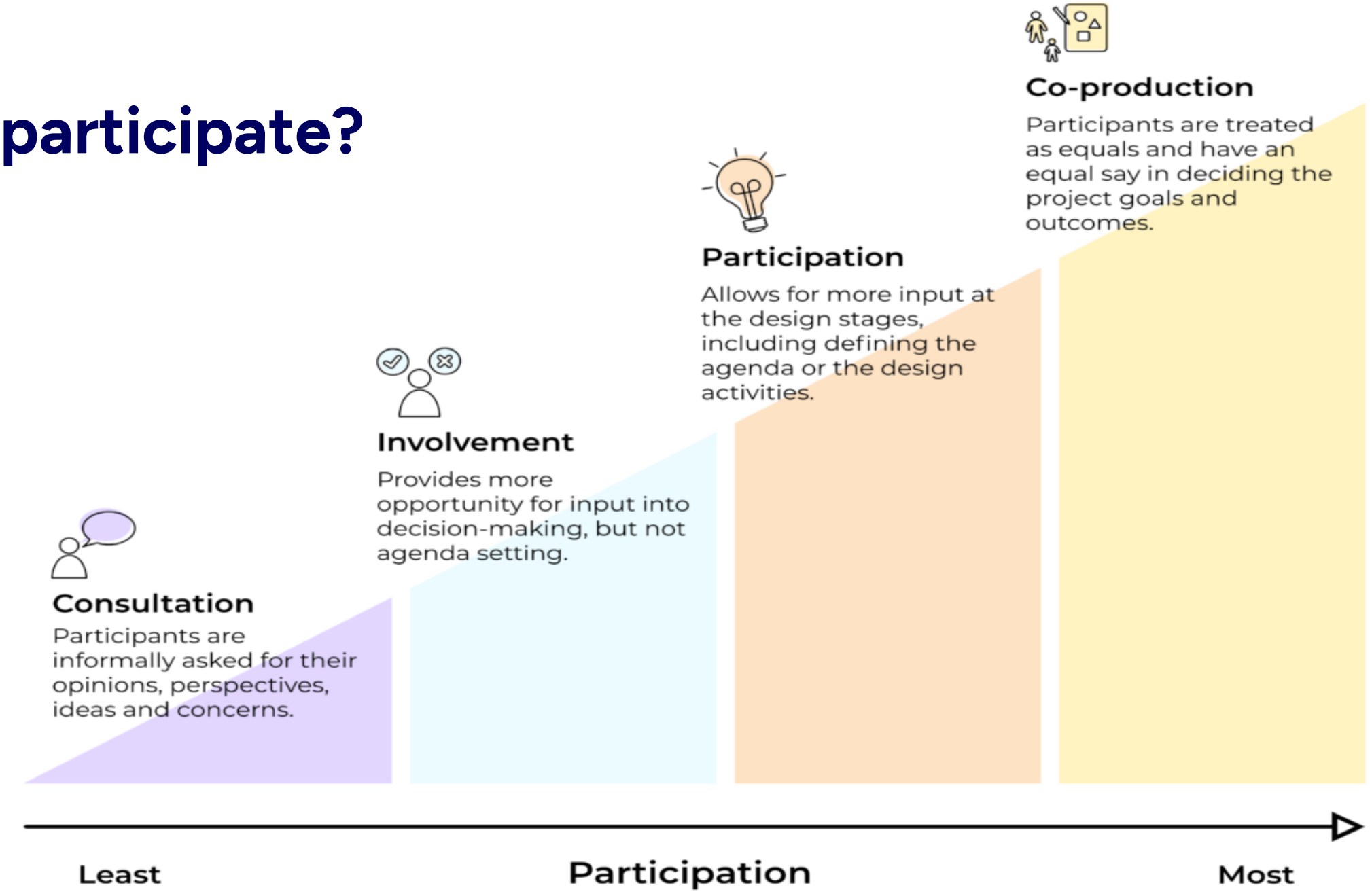
- = > A "design-implementation life-cycle model":
  - Match technology and user over multiple design cycles
  - Make technology 'generic'
  - 'designing out' too specific user representations

When participate?



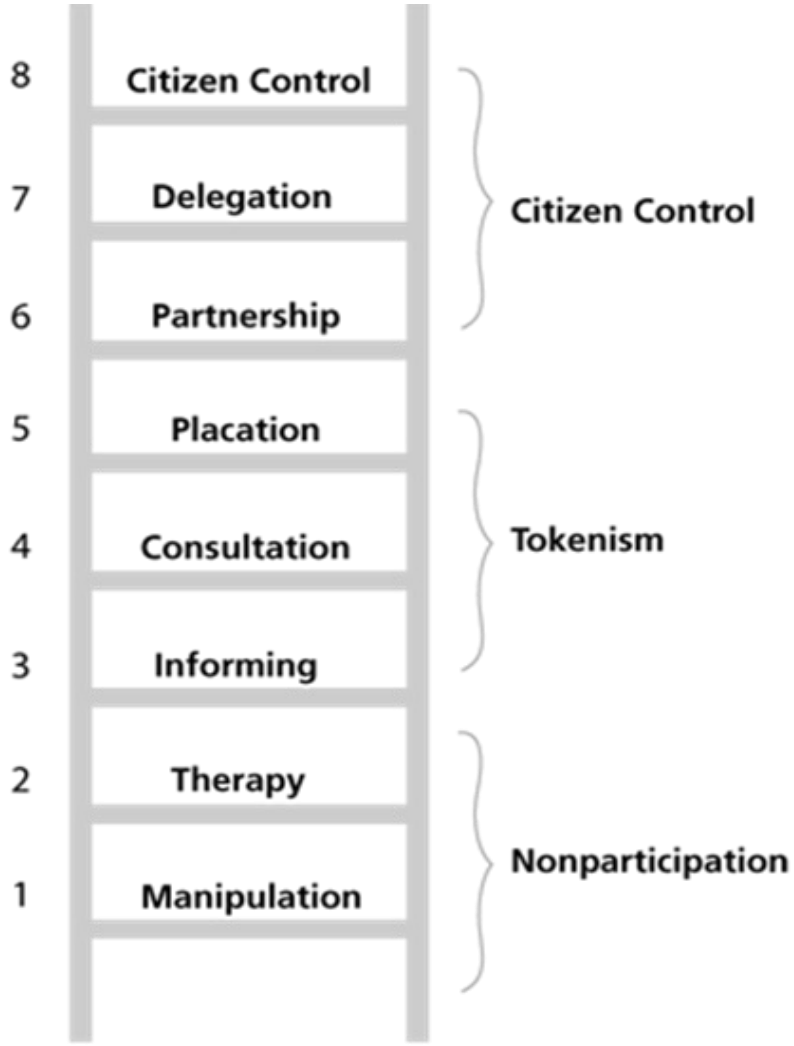
Source: Joshi and Bratteteig (2016)

When participate?



At what level?

- At a **higher** level, users can participate as equal partners, in which they have the possibility to directly influence all design decisions.
- At an **intermediate** level, users have an active role, for example as consultants providing some direct input during the design process but not always.
- At a **low level**, users are marginally involved as informants, for example by filling in questionnaires or testing prototypes in stages.
- In case of **non-participation**, the real objective is not to enable people to participate, but to enable powerholders to, control or shape the users, or "educate" or "cure" the participants



Arnstein's Ladder (1969)
Degrees of Citizen Participation

Why participate?

- Empowerment of users (Clement and van den Besselaar 1993; Namioka and Schuler 1990)
- Sense of ownership / participation itself (Robey and Farrow 1982)
- Designers better understand tacit knowledge of users (Spinuzzi 2005)
- Mutual understanding between designers and users (Simonsen and Robertson 2012)
- Better quality of products (Kensing and Blomberg 1998)
- Products better meet needs of users / higher user satisfaction (Namioka and Schuler 1990)
- Higher acceptance of products by users / increased adoption (DeLone and McLean 1992)

= > Multiple purposes, but what is ”successful” participatory design?

Whom to involve?

- Children, older people
- Relatives, friends
- Patients, caregivers, nurses
- Whose knowledge matters?
- Who are valid user representatives?

⇒ **It all depends on the premises and the purposes of user involvement (see e.g. Fischer et al. 2020)**

Björn Fischer, MSc, Alexander Peine, PhD, Britt Östlund, PhD,
The Importance of User Involvement: A Systematic Review of Involving Older Users in Technology Design, *The Gerontologist*, Volume 60, Issue 7, October 2020, Pages e513 e523,
https://doi.org/10.1093/geront/gnz163

Discussion 2 (15 min)

- In the following scenarios:
  - “Designing a campus app”
  - “Improving public transport”
  - “Redesigning a hospital waiting room”

Design a **participation strategy** by answering:

  - *when* users participate?
  - *at what level* (Arnstein’s ladder)?
  - *why* you involve them?

- https://bit.ly/3WgjWCm

